

Effect of Drug X on Outcome Y: a randomised controlled trial

1. Title and abstract

Effect of Drug X on Outcome Y: a randomised controlled trial. ^[1a]

Background: Outcome Y remains poorly controlled with standard care. **Methods:** We conducted a two-arm, parallel-group randomised trial. **Results:** The mean difference in outcome Y was 4.2 points in favour of Drug X. **Conclusions:** Drug X improves outcome Y relative to placebo. ^[1b]

2. Open science

Registered at ClinicalTrials.gov, NCT01234567, registered 2024-01-15. ^[2]

The protocol and statistical analysis plan are available at osf.io/example. ^[3]

This trial was funded by the National Research Agency (grant 42). The funder had no role in the design, conduct, or reporting of the trial. ^[5a]

There is no item 5c in CONSORT 2025 — this call exists only to exercise the “id inconnu” diagnostic. ^[5c]

3. Introduction

Prior trials suggested a plausible mechanism of action for Drug X on outcome Y. ^[6]

The primary objective was to determine whether Drug X reduces outcome Y compared with placebo, and to characterise its harms. ^[7]

4. Methods

^[11]

Adults aged 18-75 with a confirmed diagnosis of condition Z. ^[12a]

Participating sites were university hospitals with a dedicated research nurse. ^[12b]

Participants received either Drug X (10 mg once daily) or a matching placebo for 12 weeks. ^[13]

The primary outcome was change in score S from baseline to week 12, analysed as a mean difference between groups. ^[14]

Harms were monitored via routine adverse-event reporting at each study visit. ^[15]

The sample size of 240 participants was calculated to detect a 4-point difference in score S with 90% power. ^[16a]

4.1. Randomisation

The allocation sequence was generated by an independent statistician using a computer-based random-number generator. ^[17a]

Randomisation used permuted blocks of 4 and 6, stratified by site. ^[17b]

Allocation was concealed using sequentially numbered, opaque, sealed envelopes prepared by a party independent of recruitment. ^[18]

Research nurses enrolling participants had no access to the allocation sequence until after enrolment. ^[19]

Participants and outcome assessors were blinded to group assignment; care providers were not. ^[20a]

Groups were compared for the primary outcome using a mixed-effects linear model. ^[21a]

All randomised participants were analysed in their assigned group, regardless of adherence (intention-to-treat). ^[21b]

Missing outcome data were handled by multiple imputation under a missing-at-random assumption. ^[21c]

5. Results

As stated in the Methods, outcome S was assessed by a rater blinded to group assignment. ^[14]

240 participants were randomly assigned: 120 to each group. ^[22a]

Recruitment ran from March 2024 to September 2024, with follow-up completed in December 2024. ^[23a]

Table 1 shows baseline characteristics for each group. ^[25]

Characteristic	Drug X	Placebo
Age, mean (SD)	52 (12)	54 (11)
Female, n (%)	61 (51%)	58 (48%)

Table 1: Baseline characteristics.

For the primary outcome, the mean difference between groups was 4.2 points (95% CI 1.1 to 7.3). ^[26]

No serious adverse events were reported in either group. ^[27]

6. Discussion

These results are consistent with the hypothesised mechanism of action of Drug X, and are broadly in line with prior smaller trials. ^[29]

This trial was limited by its single-country setting and its 12-week follow-up, which may not capture longer-term harms. ^[30]